



Merge Sort: Todho aur Jodho!



1. Divide and Conquer Kya Hai?

UrbanPro pe padhate waqt agar ek bada sa Python ka topic samjhana ho, toh aap kya karte ho? Pura topic ek baar mein nahi samjhate! Aap usko chote-chote hisson mein **Divide (Todte)** karte ho, ek-ek hissa samjhate ho (Conquer), aur end mein sabko **Merge (Jodte)** karke puri picture clear kar dete ho.

Merge sort bilkul yahi karta hai: **Badi problem -> Choti problems -> Solve -> Merge!**



2. Step 1: Divide (Todna)

Maan lo aapke paas yeh Zomato orders ke delivery times hain (minutes mein) jinhe chote se bade kram mein lagana hai:

Numbers: 38, 27, 43, 3

Isko tab tak 2 parts (Left aur Right) mein todo jab tak har tukde mein sirf 1 number na bache (kyunki 1 number toh humesha sorted hi hota hai!).

Break 1: Do hisson mein toda.



Break 2: Fir se toda, ab sab akele hain!



3. Step 2: Merge (Jodna)

Ab hum in akele numbers ko wapas jodenge, lekin jodte waqt dhyan rakhenge ki **chota number aage aaye**.

Merge 1: 38 aur 27 ko jodo. (27 chota hai, aage rakho).



Merge 2 (Final): Ab (27, 38) aur (3, 43) ko jodo. Hum dono arrays ke pehle number ko compare karte hain (27 aur 3). 3 chota hai, toh 3 sabse aage aayega. Fir 27, fir 38, aur end mein 43.

3

27

38

43



4. Python Code for Merge Sort

Isme hume Recursion (function ke andar wahi function bulana) use karna hota hai:

```
def merge_sort(arr):
    # Agar 1 ya 0 element bacha hai, toh wo already sorted hai
    if len(arr) > 1:
        # 1. DIVIDE
        mid = len(arr) // 2      # Beech ka point nikalo
        left_half = arr[:mid]    # Left hissa
        right_half = arr[mid:]   # Right hissa

        # Left aur Right dono pe wapas merge_sort chalao
        # (Recursion)
        merge_sort(left_half)
        merge_sort(right_half)

        # 2. MERGE
        i = j = k = 0

        # Dono hisson ko compare karo aur chote wale ko 'arr'
        # mein daalo
        while i < len(left_half) and j < len(right_half):
            if left_half[i] < right_half[j]:
                arr[k] = left_half[i]
                i += 1
            else:
                arr[k] = right_half[j]
                j += 1
            k += 1

        # Agar left_half mein kuch bach gaya ho
```

```
while i < len(left_half):
    arr[k] = left_half[i]
    i += 1
    k += 1

# Agar right_half mein kuch bach gaya ho
while j < len(right_half):
    arr[k] = right_half[j]
    j += 1
    k += 1

# Test karte hain!
my_list = [38, 27, 43, 3]
merge_sort(my_list)
print("Sorted Array:", my_list)
```

🌟 Important Points to Remember!

- **Divide and Conquer:** Merge sort issi strategy pe kaam karta hai. Badi problem ko chote mein todo.
- **Recursion:** Isme function khud ko call karta hai tab tak, jab tak array ka size 1 na ho jaye.
- **Time Complexity:** Iski Time Complexity hamesha **$O(n \log n)$** hoti hai. Badi lists ke liye yeh Bubble ya Insertion sort se bahut zyada fast (tez) hota hai!
- **Extra Memory:** Isko merge karte waqt thodi extra memory chahiye hoti hai naye arrays (Left, Right) banane ke liye.

Teaching is the best way to learn! Happy Coding! 🚀